Xinsong Feng

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Research Interests

My research spans reinforcement learning, generative models, and language models. I am currently especially interested in diffusion language models, particularly in understanding their behavior and improving inference efficiency and performance. I aim to develop AI systems that are trustworthy, interpretable, and theoretically grounded.

Education

Georgia Institute of Technology *Atlanta, USA*
Ph.D. in Computer Science *Aug. 2026–Present*

University of California, Los Angeles *Los Angeles, USA*
M.S. in Electrical and Computer Engineering, GPA: 4.00/4.00 *Sep. 2023–Jun. 2025*

Chongqing University *Chongqing, China*
B.Eng. in Communication Engineering, Major GPA: 3.94/4.00 *Sep. 2019–Jun. 2023*

Research Experience

Data-Driven Decision Intelligence (D3I) Lab, William & Mary *Aug. 2025–Jun. 2026*
Research Assistant Advisor: Haipeng Chen, Assistant Professor

DELA: Learning Controllable Edits for Diffusion Language Models *Oct. 2025–Jun. 2026*

- Developed DELA, a plug-in framework that improves frozen diffusion language models through explicit draft-level editing actions: INSERT, DELETE, REMASK, and STOP.
- Designed editor and selector policies for proposing and choosing beneficial sequence edits, separating structural control from token-level denoising.
- Designed a curriculum learning strategy that combines inverse-edit supervision with rollout-guided preference learning for effective edit proposal and selection.
- Built the full experimental pipeline for candidate generation, verifier-based selection, and benchmark evaluation under a fixed-backbone setting.

Constrained Optimization via RL and ODE-based Models *Mar. 2025–Sep. 2025*

- Proposed Constrained Markov Flow Optimizer (CMFO), a framework unifying reinforcement learning and ODE-based generative models for constrained optimization.
- Formulated constrained optimization as sequential decision-making over iterative refinement dynamics, establishing a connection between policy learning and ODE-based optimization.
- Introduced RelaxCompletion for differentiable feasibility handling and achieved state-of-the-art performance with substantially faster inference than classical solvers.

University of California, Los Angeles *Sep. 2023–Jun. 2025*
M.S. Student Researcher Research collaboration with Prof. Haipeng Chen

Offline Reinforcement Learning with Generative Trajectory Policies *Aug. 2024–May 2025*

- Proposed Generative Trajectory Policies (GTP), a unified offline RL framework connecting diffusion, flow matching, and consistency models through a continuous-time trajectory formulation.
- Derived a closed-form score approximation from offline data, eliminating iterative ODE solvers and enabling stable and efficient policy learning.
- Introduced advantage-weighted value guidance and achieved state-of-the-art results on D4RL, including perfect scores on multiple AntMaze tasks.

Sequential Stochastic Combinatorial Optimization with Hierarchical RL *Oct. 2023–May 2024*

- Formulated sequential stochastic combinatorial optimization as a new decision-making setting with uncertain dynamics and combinatorial action spaces.
- Developed a hierarchical RL framework with tailored state, reward, and null-action design for efficient learning over large combinatorial action spaces.
- Proposed a wake-sleep option training strategy and outperformed strong baselines by 20% on adaptive influence maximization and route planning.

Publications

[1] H. Meng, **X. Feng**, et al. "Solving Constrained Optimization Problems as ODE-based Models Using Reinforcement Learning," *TMLR*, 2026.

[2] **X. Feng**, L. Tang, C. Wang, and H. Chen. "Offline Reinforcement Learning with Generative Trajectory Policies," *ICML*, 2026. *arXiv preprint* arXiv:2510.11499, 2025.

[3] **X. Feng**, Z. Yu, Y. Xiong, and H. Chen. "Sequential Stochastic Combinatorial Optimization Using Hierarchical Reinforcement Learning," *ICLR*, 2025. *arXiv preprint* arXiv:2502.05537, 2025.

[4] B. Zhou, X. Guo, **X. Feng**, et al. "Complex-Frequency Selective Resonances in Disordered Time-Varying Cavities," *Laser & Photonics Reviews*, e01228, 2025.

[5] **X. Feng**, I. Roberts. "Adaptive Cell Range Expansion in Multi-Band UAV Communication Networks," *arXiv preprint* arXiv:2411.18123, 2024.

[6] B. Zhou, **X. Feng**, et al. "Transition from normal to non-normal distributions of an electromagnetic field in a disordered time-varying cavity," *Physical Review A*, 110(6), 063524, 2024.

[7] B. Zhou, **X. Feng**, et al. "Susceptibility Invariance and Duality-Matching Condition for Perpendicular-Motion Metasurface," *Advanced Physics Research*, 3(10), 2400073, 2024.

[8] Y. Wei, L. Liang, B. Zhou, and **X. Feng**. "A Modified Blockchain DPoS Consensus Algorithm Based on Anomaly Detection and Reward-Punishment," *IEEE International Conference on Communication Software and Networks*, 283–288, 2021.

Teaching Experience

Graduate Teaching Assistant, *University of California, Los Angeles* *Spring 2024*

Course: *Introduction to Communication Systems (ECE 132A)*. Graded assignments and exams, provided feedback to improve students' understanding, and worked with the instructor to maintain consistent grading standards.

Undergraduate Teaching Assistant, *Chongqing University* *Fall 2021–Fall 2022*

Courses: *Digital Signal Processing* (Fall 2021, Fall 2022); *Introduction to Programming with C++* (Fall 2022). Prepared course materials, supported lectures and office hours, graded homework and final exams, and was recognized as a "Top 10 Outstanding Teaching Assistant."

Selected Honors and Awards

Distinguished Student, Chongqing University (top 10/266 in department) *May 2023*

Top 10 Outstanding Teaching Assistant, Chongqing University *Jan. 2023*

National Scholarship for Academic Excellence (top 4/266 in department) *Dec. 2022*

Outstanding Graduate, Chongqing University (top 25/266 in department) *Nov. 2022*

First Prize, Chinese Mathematics Competitions (twice; top 0.2% nationally) *Dec. 2020, Dec. 2021*

Second Prize, Undergraduate Mathematical Contest in Modeling (top 2.3% nationally) *Nov. 2021*

Professional Service

Reviewer, NeurIPS 2026 *Jun. 2026*

Reviewer, AISTATS 2026 *Oct. 2025*

Reviewer, IEEE GLOBECOM 2024 (Signal Processing for Communications Symposium) *Jun. 2024*

Skills

Programming: Python, C++, Java, MATLAB

ML / Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, OpenAI Gym

Generative AI & RL: Offline RL, Policy Optimization, Diffusion Models, LLM Fine-tuning, Model Evaluation

Systems & Tools: Linux, Git, Docker, Kubernetes